Exam 4.

*You may use a basic calculator and formula sheets. Your sheets must contain formulas only, no examples. Phones and all other smart devices are forbidden. Any AI usage will result in an automatic F for the exam. Show **ALL** your work for credits.*

Notice: For all the maximization problems, it is required to do the three-step process to solve the problem.

1. Find $f'(x)$. Simplify 1c.

a. $f(x) = 5x^6 - 7x^2 + 9x + 2026$

$f'(x) = (5x^6 - 7x^2 + 9x + 2026)'$

$= \boxed{30x^5 - 14x + 9}$

b. $f(x) = \frac{3}{x^3} + \frac{4}{x^4} + \frac{5}{x^5} + 6x + 7$

$f(x) = 3x^{-3} + 4x^{-4} + 5x^{-5} + 6x + 7$

$f'(x) = (3x^{-3} + 4x^{-4} + 5x^{-5} + 6x + 7)'$

$= \boxed{-9x^{-4} - 16x^{-5} - 25x^{-6} + 6}$

c. $f(x) = \frac{x+2026}{x-2026}$ (Simplify the answer)

$f'(x) = \frac{(x+2026)' \cdot (x-2026) - (x+2026)(x-2026)'}{(x-2026)^2}$

$= \frac{1(x-2026) - (x+2026)1}{(x-2026)^2} = \frac{x - 2026 - x - 2026}{(x-2026)^2} = \boxed{\frac{-4052}{(x-2026)^2}}$

d. $f(x) = (x^2 + 1)(x^5 + 2)$

$f'(x) = (x^2+1)' \cdot (x^5+2) + (x^2+1)(x^5+2)'$

$= \boxed{2x(x^5+2) + (x^2+1) \cdot 5x^4}$

Each Question (part) is worth 10 points.

2. The per-day cost function of the manufacture

$$C(x) = 25{,}00 + 10x + x^2$$

where x is the number of MP3 players manufactured per day. Assume that the company cannot manufacture more than 100 MP3 players per day (x is no more than 100).

a. Find the marginal Cost function.

$MC = C'(x) = (2500 + 10x + x^2)' = \boxed{10 + 2x}$

b. Interpret the meaning of C' (30).

$C'(30) = 10 + 2 \times 30 = \boxed{600}$

$C'(30) \approx$ cost of producing the 31st item

c. Write the average cost function

$$AC = \frac{2500 + 10x + x^2}{x}$$

$$= \frac{2{,}500}{x} + 10 + x$$

d. How many items should be manufactured in order to minimize the average cost?

$f'(x) = (2{,}500x^{-1} + 10 + x)'$

$= -2{,}500x^{-2} + 1$

$0 = \frac{-2{,}500}{x^2} + 1$

$\frac{2{,}500}{x^2} = 1$

$x^2 = 2500$

$x = \pm 50$

items must be positive

$\rightarrow x = 50$

$f''(x) = 5000x^{-3}$

$f''(50) = \frac{5000}{50^3} = 0.04 > 0$

↳ Confirm that $x = 50$ minimizes the average cost

...ch manufacturer has determined that the daily price p (in dollars) they can charge is related to the quantity x they produce by the following quadratic demand function.

$$p(x) = -x^2 + 15x + 1800$$

a. Find the revenue function and the marginal revenue (MR) function

$R(x) = (-x^2 + 15x + 1800)x$

$= -x^3 + 15x^2 + 1800x$

$MR(x) = (-x^3 + 15x^2 + 1800x)'$

$= -3x^2 + 30x + 1800$

b. Interpret the meaning of MR (10).

$MR(10) = -3 \times 10^2 + 30 \times 10 + 1800 = 1800$

$MR(10) \approx$ revenue of selling the 11th item

c. Find the quantity to maximize the revenue function

$R'(x) = -3x^2 + 30x + 1800$

$0 = -3x^2 + 30x + 1800$

$0 = -3(x^2 - 10x - 600)$

$0 = x^2 - 10x - 600$

$0 = (x - 30)(x + 20)$

$x = 30 ; x = -20$

Item must be positive

$\rightarrow x = 30$

$R''(x) = (-3x^2 + 30x + 1800)'$

$= -6x + 30$

$R''(30) = -6 \times 30 + 30$

$= -150 < 0$

$\rightarrow$ Confirm that $x = 30$ maximizes the revenue function

4. A company determines that when x units of a product are produced each month, they will be sold at the price of

$$p = 80 - x$$

dollars per unit. The total cost of producing the x units will be

$$C(x) = x^2 + 20x + 150$$

a. Find the profit function and the marginal profit (MP) function

$P(x) = (80 - x)x - (x^2 + 20x + 150)$

$= 80x - x^2 - x^2 - 20x - 150$

$= -2x^2 + 60x - 150$

$MP = (-2x^2 + 60x - 150)'$

$= -4x + 60$

b. Interpret the meaning of MP(5).

$MP(5) = -4 \times 5 + 60 = 40$

$MP(5) \approx$ profit of selling the 6th item

c. How many units should the company produce to maximize the profit?

$P'(x) = -4x + 60$

$0 = -4x + 60$

$4x = 60$

$x = 15$

$P''(x) = (-4x + 60)'$

$= -4$

$P''(15) < 0$

↳ Confirm that $x = 15$ maximizes the profit